GRAVITY RACER PERFORMANCE

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Introduction

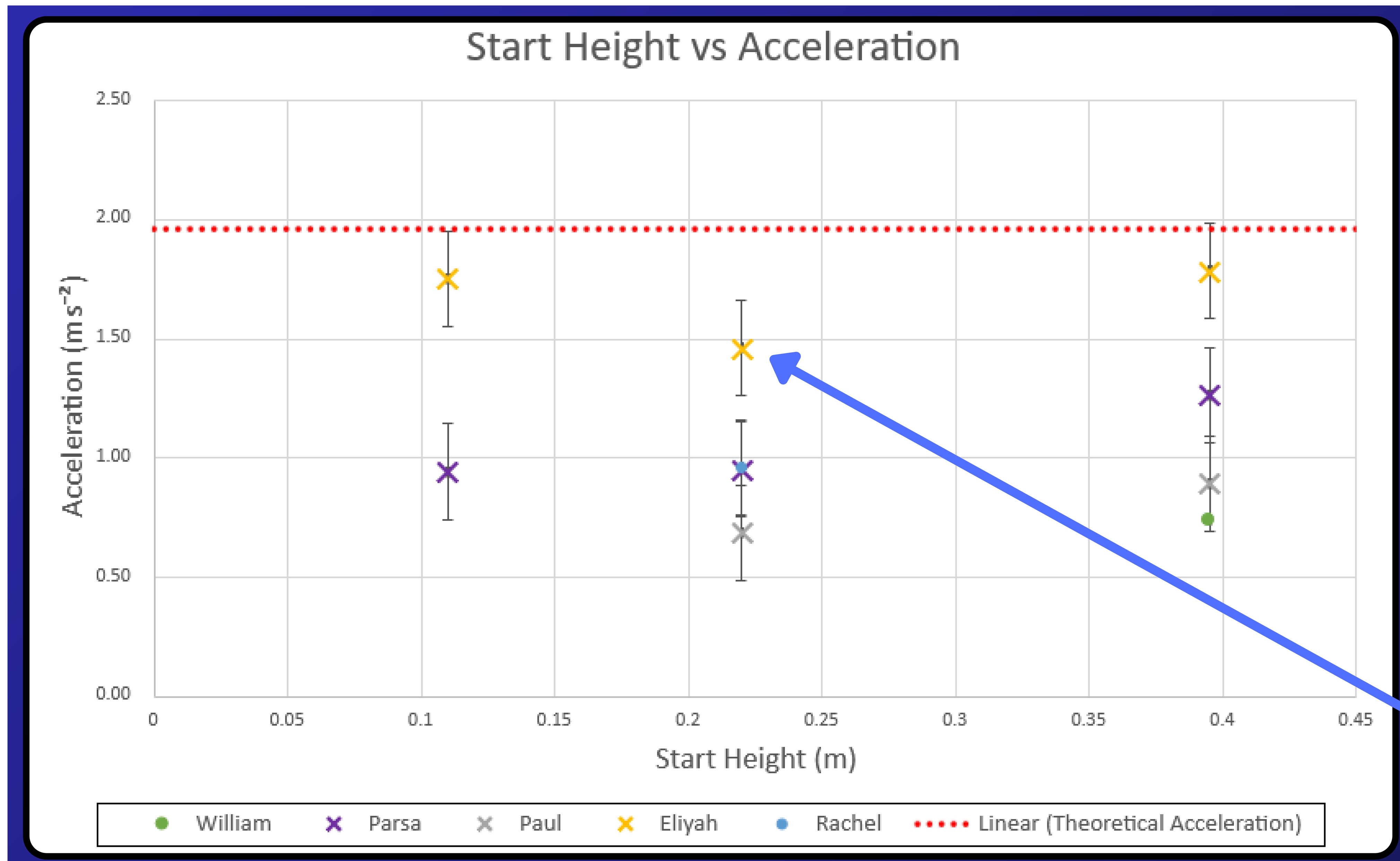
This following poster outlines how different gravity racers performed against theoretical predictions. Through various experiments and testing methods, different parameters were measured and recorded, which were later plotted to determine key values such as acceleration and velocity. The testing of the gravity racers allowed us to make connections into how different design elements can affect the performance in different ways, by considering elements such as mass, shape and material. This poster takes you through their performance under different conditions and how it relates to theoretical estimations.

Methods

The gravity racer's performance was tested on an adjustable ramp, fixed at 19.5° to the horizontal. To investigate the effect of height, the start height was varied while mass remained constant. Again, the racer was released from rest, and time was measured over a set distance. Average times were used to calculate experimental velocity and acceleration. These results were compared to theoretical predictions based on conservation of energy and inclined plane motion. All data was plotted and analysed to evaluate how closely the experimental values matched theoretical expectations.

To isolate the effect of mass, additional weights were added while keeping the ramp angle and travel distance constant. The racer was released from rest for each mass, and the travel time was recorded with a stopwatch over three trials to calculate an average. Acceleration was then derived using kinematic equations.

Data



Graph 1: Start Height vs Acceleration

Height Uncertainty δH (m)	0.0005
Stopwatch Uncertainty δS (s)	0.0005
Reaction Time δR (s)	0.2000
Displacement Uncertainty δD (m)	0.0005
Acceleration [Height] δ1 (m/s2)	0.2000
Acceleration [Mass] δ2 (m/s2)	0.0007

Table 1: Uncertainty for both experiments

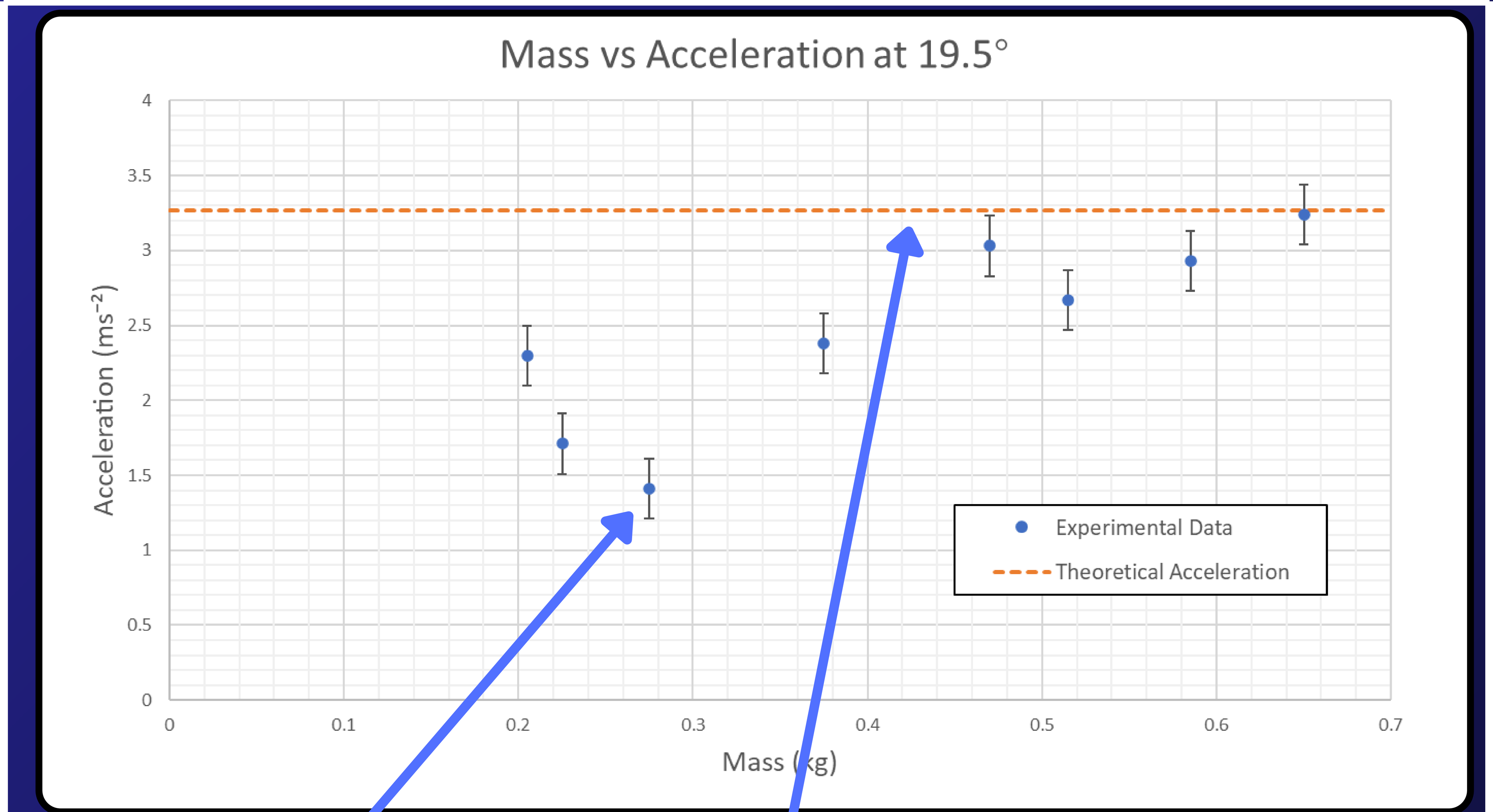
$$\delta 1 = \sqrt{\delta S^2 + \delta R^2 + \delta D^2 + \delta H^2}$$

Total Uncertainty for Height Trials

$$\delta 2 = \sqrt{\delta S^2 + \delta R^2 + \delta D^2}$$

Total Uncertainty for Mass Trials

All of the calculated accelerations fall below the theoretical acceleration, with Eliyah's being closest (15.98% mean difference). All of the calculated accelerations are relatively linear, as expected, with deviations from the theoretical value beign attributed to frictional effects between the wheels, ramp, and internal components of the racer. The significance of these effects varied between the gravity racers, which is reflected in the data



Graph 2: Mass vs Acceleration

None of the experimental accelerations overlap with the theoretical acceleration, apart from the greatest value of mass, which lies extremely close to the expected value, with a percentage difference of 0.92%

The calculated acceleration values showed significant deviation from the theoretical value at lower masses, but progressively approached the theoretical acceleration of 3.27 m/s² as mass increased

$t = \sqrt{\frac{2s}{a}}$ Theoretical travel time

$t = 1.05s$ Theoretical minimum travel time

$a = g \sin\theta = 3.27ms^{-1}$ Theoretical maximum acceleration

Discussion

For the varying mass experiment, the results had large deviations from the theoretical predictions. The theoretical acceleration stayed constant at 3.27m/s², which is expected due to the mass being an independent variable. However, the experimental results had a range of accelerations from 1.14 m/s² to 3.24m/s², none of which reaching the theoretical value, due to the energy losses not being accounted for in the ideal model. The larger masses produced higher accelerations closer to the theoretical prediction, whereas the lower masses had a greater deviation. This suggests that air drag and friction affected the smaller masses more.

For the varying height experiment, theoretically for an increase in angle the final velocity should increase, but acceleration stays the same for a fixed angle. There were big time differences for the repeated results. This was due to different people releasing the gravity racer and measuring the time, and human reaction time causing an uncertainty.

Overall, both the mass and the height experiment show the correct trends, however, they have discrepancies due to frictional loses and measurement uncertainties, which aren't taken into account in the ideal theoretical model.

Conclusion

Experimental trials compared multiple handmade gravity racers against theoretical predictions across varying release heights and car masses.

For height variance, the data fell short of theoretical expectations with a minimum deviation of 15.98%. While each individual racer demonstrated internal consistency across trials, the clear performance gaps between the different vehicles highlight that manufacturing quality is a decisive factor not captured by the model.

For mass variation, the collective results from the racers correlated poorly with the model, yielding an average deviation of 74.2% with small uncertainties. Theory predicted constant performance, but the experimental results revealed a linear upward trend across the trials.

Driscoll, H.F., Bullas, A.M., King, C.E., Senior, T., Haake, S.J. and Hart, J., 2016. Application of Newtonian physics to predict the speed of a gravity racer. Physics Education, 51(4), p.045002